INCH-POUND
MIL-DTL-24704C(SH)
31 July 2019
SUPERSEDING
MIL-DTL-24704B(SH)
4 October 2011

DETAIL SPECIFICATION

FLANGES, FOUR-BOLT, SQUARE, FOR HYDRAULIC SYSTEMS, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for four-bolt square flanges for use in 3,000 pounds per square inch (lab/in²) hydraulic systems.

1.2 Classification. Flanges are designated by a part or identifying number (PIN), as specified in the applicable specification sheet (see 6.2).

1.2.1 Material requirement. The material requirement is designated by a letter. The material is designated in the applicable specification sheet.

1.2.2 Vent requirement. The vent requirement is designated by a letter. The letter designation and location for the vent port on a given flange is in accordance with the applicable specification sheet.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-24704/1	-	Flanges, Four-Bolt Square, Socket Weld, for Hydraulic Systems
MIL-DTL-24704/2	-	Flanges, Four-Bolt Square, Butt Weld, for Hydraulic Systems
MIL-DTL-24704/4	-	Flanges, Four-Bolt Square, Blind, for Hydraulic Systems
MIL-DTL-24704/5	-	Insert Sleeves, for Four-Bolt Flanges, for Hydraulic Systems

Comments, suggestions, or questions on this document should be addressed to: Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 4810

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM E1282 - Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Materials. Flanges shall be manufactured from the materials designated in the applicable specification sheet.

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Machining. Flanges shall be machined to the configuration and dimensions as specified herein and on the applicable specification sheet.

3.4 Surface finish. Unless otherwise specified (see 6.2), surface roughness shall not exceed 125 microinches when inspected as specified in 4.1.2. Surface finish for sealing surface shall be in accordance with the applicable specification sheet. Machined surfaces shall have no burrs. Sharp edges and corners shall be rounded to a radius of 0.005 inch to 0.015 inch. Non-machined surfaces, such as forged surfaces and bar stock flats, shall have no fins, cracks, or spongy areas.

3.5 Identification marking. Each flange shall be clearly and permanently marked with the following information:

a. PIN (see 1.2).

b. Manufacturer’s name, trademark, or logo.

c. Manufacturer’s part number (optional).

d. Pressure rating.

3.6 Workmanship. Flanges shall be free from detrimental grooving, indentations, flaws, cracks, lamination, surface pitting, inclusions, and slivers that would interfere in their use and shall be free from dirt, corrosion, and other extraneous material.

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4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall consist of the examinations and inspections specified in table I.

TABLE I. Classification of defects.

Categories	Defects	Inspection method
Major		
101	Surface finish not as specified on sealing face (see 3.4)	4.1.2
102	Flange size not as specified (see 3.3)	SIE 1/
103	Sealing face not machined as specified (see 3.3)	SIE
104	Bore dimensions not as specified (see 3.3)	SIE
105	Bolt holes not located or drilled properly (see 3.3)	SIE
106	Vent location not as specified (see 3.3)	SIE
107	Material composition not as specified (see 3.2)	ASTM E1282
108	Flange style not as specified (see 3.3)	SIE
109	Workmanship defects (see 3.6)	Visual
Minor		
201	Surface finish not as specified (not including sealing face) (see 3.4)	4.1.2
202	Flange not properly marked (see 3.5)	Visual

NOTE:
1/ Standard inspection equipment (SIE).

4.1.1 Sampling. Unless otherwise specified (see 6.2), sampling for conformance inspection shall be as specified in 4.1.1.1 through 4.1.1.3.

4.1.1.1 Inspection lot. A lot shall consist of finished flanges that are of the same material, type, and size; fabricated by the same process; produced as one continuous run or order, or part thereof; and submitted for acceptance inspection at the same time.

4.1.1.2 Sampling for examination. A random sample of flanges shall be selected from each lot and inspected in accordance with table I. The sampling plans in table II shall be utilized in accordance with the classification of defects in table I.

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TABLE II. Sampling plan for major and minor defects.

Lot size	Sample size
1 to 13	Entire lot
14 to 150	13
151 to 280	20
281 to 500	29
501 to 1,200	34
1,201 to 3,200	42

NOTES:
1. All defective items shall be replaced with acceptable items prior to lot acceptance.
2. Stop inspecting samples when rejection criteria are reached (see 4.1.1.3).

4.1.1.3 Rejection. Any flange in the sample that contains one or more defects and does not meet the requirements in table I or the applicable specification sheets shall be rejected. If there are any number of defective flanges in any sample, the entire lot shall be rejected.

4.1.2. Surface finish inspection method. Surface roughness shall be identified in accordance with ASME B46.1. The determination of surface finishes shall be made with a profilometer, comparator brush analyzer, or other comparison equipment with an accuracy of ±15 percent at the level being measured. Surface finishes on the sealing face not in accordance with this specification (see 3.4) or the applicable specification sheet shall be cause for rejection.

4.1.3 Material composition. When specified (see 6.2), a chemical analysis shall be performed in accordance with ASTM E1282 for verification of material composition.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Flanges covered by this specification are intended for use in 3,000 lb/in² hydraulic systems that operate under conditions that impose mechanically-induced shock on system piping and components.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Title, number, and date of applicable specification sheet.
c. Specification sheet PIN required (see 1.2).
d. Whether surface roughness measurements are other than specified (see 3.4).
e. Whether conformance sampling is other than specified (see 4.1.1).
f. Whether material verification by chemical analysis is required (see 4.1.3).

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g. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Flange, right angle

Flange, straight

Piping

Tubing

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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FLANGE INTERFACE AND ASSEMBLY INSTRUCTIONS

A.1 SCOPE

A.1.1 Scope. This appendix details the interface machining configuration and assembly instructions for four-bolt flanges. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

A.2 APPLICABLE DOCUMENTS

A.2.1 General. The documents listed in this section are specified in sections 3 or 4 of this appendix. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this appendix, whether or not they are listed.

A.2.2 Government documents.

A.2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-59004 - Anti-Galling Compound, Thread Lubricating, Seizing Resistant, and Calcium Hydroxide Containing

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-45932 - Insert, Screw Thread, Thin Wall, Locked in: General Specification for

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.2.3 Non-Government publications. The following documents form a part of this appendix to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS1351 - Screw, Cap, Socket Head, Undrilled and Drilled, Plain and Self-Locking, Alloy Steel, Corrosion-Resistant Steel and Heat-Resistant Steel, UNRF-3A

NASM7838 - Bolt, Internal Wrenching, 160 KSI Ftu

NASM8846 - Insert, Screw-Thread, Helical Coil

(Copies of these documents are available online at www.aia-aerospace.org.)

SAE INTERNATIONAL

SAE-AS3209 - Packing, Preformed - AMS7276, 'O' Ring

(Copies of this document are available online at www.sae.org.)

A.2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this appendix and the references cited herein (except for related specification sheets), the text of this appendix takes precedence. Nothing in this appendix, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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A.3 INTERFACE CONNECTION

A.3.1 Interface connection. The interface configurations for cavities for the flanges covered by this specification are identified on figure A-1.

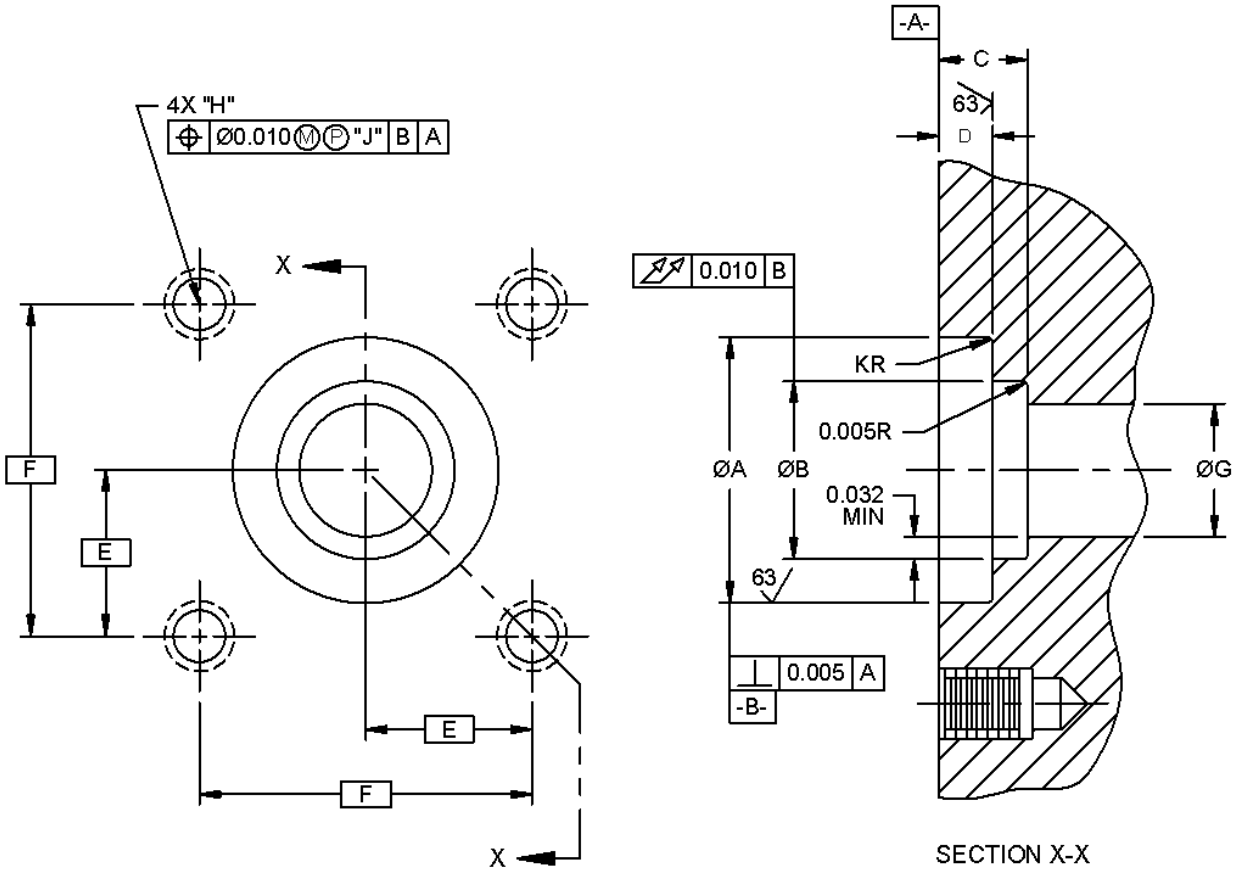


FIGURE A-1. Machining dimensions.

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Flange code	Flange size (square)	A +0.002 -0.000	B ±0.001	C +0.010 -0.000	D ±0.003	E	F	G Max	H Thread size	J Min	K ±0.005
A	1.50	0.488	0.279	0.251	0.118	0.469	0.938	0.215	0.250-28UNF	0.500	0.010
B	1.75	0.688	0.439	0.251	0.153	0.562	1.125	0.375	0.312-24UNF	0.500	0.010
C	2.00	1.000	0.689	0.314	0.184	0.656	1.312	0.625	0.375-24UNF	0.500	0.010
D	2.25	1.188	0.876	0.314	0.184	0.750	1.500	0.812	0.438-20UNF	0.750	0.010
E	2.75	1.438	1.064	0.377	0.247	0.938	1.875	1.000	0.438-20UNF	0.750	0.010
F	3.50	2.125	1.564	0.502	0.311	1.250	2.500	1.500	0.500-20UNF	1.250	0.010
G	4.25	2.375	1.814	0.502	0.311	1.531	3.062	1.688	0.625-18UNF	1.625	0.010
H	5.75	3.000	2.445	0.502	0.311	2.000	4.000	2.125	0.750-16UNF	2.125	0.010

NOTE:

1. Unless otherwise specified herein:

a. Surface roughness should not exceed 125 microinches.

b. Tolerances should be:

(1) X.XXX: ±0.015

(2) X.XX: ±0.03

FIGURE A-1. Machining dimensions – Continued.

A.3.1.1 Bolt thread engagement for steel. The recommended bolt thread engagement for steel-based materials that do not require threaded inserts is approximately 1.5 times the bolt diameter unless otherwise required by engineering analysis or the applicable specification.

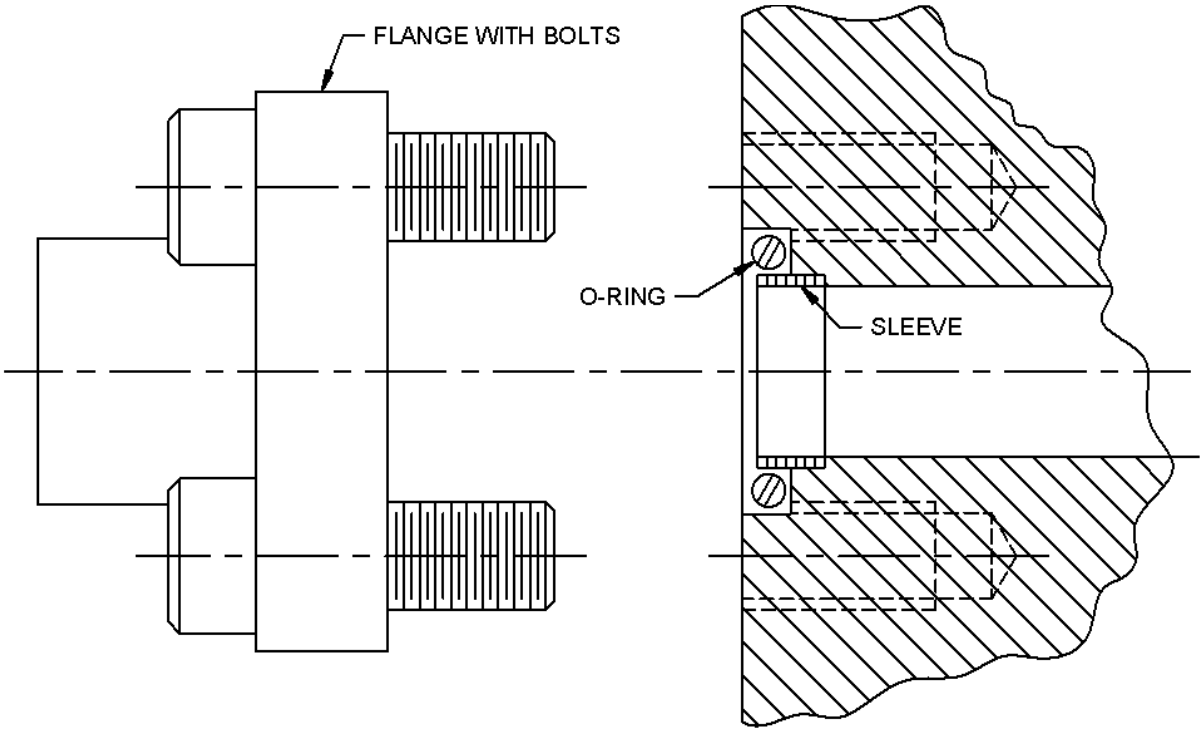
A.3.1.2 Bolt thread engagement for aluminum. Guidance for determining bolt thread engagement for aluminum-alloy-based materials that require threaded inserts is provided by NASM8846, for standard inserts, or MIL-DTL-45932, for thin-wall inserts. However, if lower torque values can be utilized because the full tensile strength of the fastener is not required for a specific application, it is not necessary to use the longer length threaded inserts that would be required for use with the higher tensile strength and higher torque values. As a minimum, the insert length selected should not result in tensile failure in parent or tapped material assuming a tensile load that is 135 percent of the worst-case in-service load on the insert. The in-service load on the insert may be due to a combination of the fastener being tightened to an applicable assembly drawing torque value and the application of system pressure or other external loading on the flange.

A.4 ASSEMBLY INSTRUCTIONS

A.4.1 Assembly instructions. Assembly should be as shown on figure A-2. The selection of the O-ring and sleeve should be made from figure A-2.

A.4.2 Fasteners. The recommended fasteners are socket-head cap screws made of heat-resistant steel (UNS S66286) in accordance with NAS1351 or internal wrenching bolts in accordance with NASM7838. The length of the cap screw is determined by adding the thickness of the flange and washer (if required) to the minimum thread engagement, as determined in A.3.1.1 or A.3.1.2, then selecting the nearest standard cap screw size. Fasteners should be safety wired or lock wired and lubricated using a general-purpose thread lubricant, such as A-A-59004.

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Flange code 1/	Flange size	O-ring SAE-AS3209	Sleeve code 2/
A	1.50	AS3209-012	A
B	1.75	AS3209-112	B
C	2.00	AS3209-210	C
D	2.25	AS3209-213	D
E	2.75	AS3209-217	E
F	3.50	AS3209-327	F
G	4.25	AS3209-329	G
H	5.75	AS3209-334	H

NOTES:
1/ From figure A-1.
2/ From MIL-DTL-24704/5.

FIGURE A-2. O-ring and sleeve selection guide.

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CONCLUDING MATERIAL

Preparing activity:
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